

JEE MAIN REVISION MASTER: ATOMIC STRUCTURE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 12 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	de Broglie, Heisenberg & Quantum Rules (Aufbau, Hund, Pauli)	8 PYQs	Quantum Mechanics
Moderate	Bohr's Atomic Theory & Hydrogen Spectral Lines	4 PYQs	Atomic Models

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
de Broglie, Heisenberg & Quantum Rules (Aufbau, Hund, Pauli)	$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$, $\mu = \sqrt{n(n+1)} \hbar$, $\text{Max electrons in subshell} = 2(2l+1)$
Bohr's Atomic Theory & Hydrogen Spectral Lines	$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$, $\bar{\nu} = 109677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ cm}^{-1}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 de Broglie, Heisenberg & Quantum Rules (Aufbau, Hund, Pauli) (High Yield — 8 PYQs)

Core Principle: Heisenberg's uncertainty principle: $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$. Quantum numbers: n (size/energy), l (shape: $s=0, p=1, d=2, f=3$), m_l (orientation, $-l$ to $+l$), m_s (spin $\pm 1/2$). Aufbau rule fills orbitals in order of increasing $(n+l)$. Pauli exclusion principle: no two electrons can have all 4 quantum numbers identical. Hund's rule: pairing occurs only after every degenerate orbital has one electron with parallel spin.

■ **Caution / JEE Pitfall:** When $(n+l)$ values are equal, the orbital with LOWER n has lower energy and fills first (e.g. 3d with $n+l=5$ vs 4p with $n+l=5$: 3d fills before 4p).

■ **Speed Trick:** Chromium ($Z=24$: $[\text{Ar}] 3d^5 4s^1$) and Copper ($Z=29$: $[\text{Ar}] 3d^{10} 4s^1$) have anomalous configurations due to extra stability of half-filled and completely-filled d-subshells.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Which of the following is the CORRECT statement about Ψ^2 ?..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Which of the following represents de Broglie equation?..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "According to the Heisenberg's Uncertainty principle, the value of $\Delta b \cdot \Delta x$ for an object whose mass is 10^{-6} kg is $(h=6.626 \dots)$ " [Ans: N/A]

3.2 Bohr's Atomic Theory & Hydrogen Spectral Lines (Moderate Yield — 4 PYQs)

Core Principle: Bohr model: $r_n = 0.529 (n^2 / Z) \text{ \AA}$, $E_n = -13.6(Z^2 / n^2) \text{ eV}$. Wave number $\bar{\nu} = (1 / \lambda) = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$. Lyman series is in UV region, Balmer series is in Visible region.

■■ **Caution / JEE Pitfall:** Bohr theory is applicable ONLY to single-electron species (extH, extHe⁺, extLi²⁺, extBe³⁺); it fails for multi-electron atoms.

■ **Speed Trick:** Balmer series: The only hydrogen spectral series visible to the human eye ($n_1 = 2$).

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): "The number of orbitals associated with 'N' shell of an atom is..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "The energy associated with first orbit of He⁺ is..." [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): "If wavelength of photon is $2.2 \times 10^{-11} \text{ m}$ and $E = 6.6 \times 10^{-34} \text{ Js}$, then momentum of photon..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Which of the following is the CORRECT statement about Ψ^2 ?

- (A) Ψ^2 represents atomic orbit
- (B) Probability density of the electron at that point
- (C) $\Psi^2 \neq 0$ for nodes
- (D) Ψ^2 has no physical meaning

Correct Answer: None

Solution Analysis: The correct statement is: Option B: Probability density of the electron at that point. In quantum mechanics, Ψ is the wave function of an electron. The wave function itself, Ψ , does not have direct physical meaning. But the square of the wave function gives the probability density of finding the electron at a particular point. More correctly, we write it as: $|\Psi|^2$. For real wave functions, this is often written simply as: Ψ^2 . So, Ψ^2 tells us the probability density of the electron at that point. Also, at nodes, the probability of finding the electron is zero, so:

Benchmark Problem 2: KCET 2025 (Q#3)

The number of orbitals associated with 'N' shell of an atom is

- (A) 16
- (B) 32
- (C) 3
- (D) 4

Correct Answer: None

Solution Analysis: The "N" shell corresponds to principal quantum number $n = 4$. The total number of orbitals in a shell is given by n^2 . Here, $n=4$. So, number of orbitals = $4^2=16$. Answer: Option A (16)